

SkillScreen is an end-to-end automated interviewing platform that generates role-tailored interviews from resumes, job descriptions, or recruiter-specified skills — delivered as chat, audio, or video interviews. It scores hard and soft skills, detects cheating risks, and produces auditable evidence (clips, transcripts, test results) that recruiters and hiring teams can act on.

1) Objective and End-Goal

Objective: Create a reliable, explainable, enterprise-grade interview system that reduces manual interviewer effort, increases hiring signal quality, and accelerates screening at scale.

End-Goal: A commercially viable product that recruiters trust for screening and shortlisting: accurate automated scoring (hard + soft skills), low false-positive proctoring flags, seamless ATS integration, and clear evidence for every automated judgment.

2) Potential User Personas

- **Recruiter (Core Buyer):** Needs fast, consistent screening and clear evidence to shortlist at scale.
- **Hiring Manager:** Wants high-signal hard-skill assessments and short evidence clips to justify passes.
- **Candidate:** Wants clear instructions, fair scoring, and actionable feedback.
- **Technical Evaluator / Team Lead:** Uses detailed code artifacts and runtime results to validate technical fit.
- **Compliance Officer / HR Ops:** Needs retention controls, audit logs, and bias/fairness reporting.

3) Value Proposition

SkillScreen saves interviewers' time by automating repeatable screening work, raises signal quality by combining multimodal assessments (code + speech + video), and provides tamper-evident evidence for hiring decisions. It's built for fairness and auditability: every score is traceable to clips, code tests, and transcripts so humans can always override.

4) Features

Orchestration & Profiles

- Resume parsing (skill extraction, experience timeline).
- Job-description ingestion and skill mapping → auto-generate role-weighted question sets.
- Interview templates (behavioral, technical, system design, live-coding, case study).
- Candidate onboarding checks: device, mic, camera, network, room privacy consent.
- Timezone-aware scheduling & calendar integration (Google / Outlook).

Interaction Modes

- Chat-based interview (text prompts + code snippets).
- Audio-only interview (streaming voice).
- Video interview (webcam + optional screen-share).
- Hybrid: text + audio + video per question.

Recording & Artifacts

- Synchronized multi-track capture: webcam, mic, screen, chat, and code.
- High-quality batch processing: post-meeting ASR polishing, speaker alignment.
- Raw artifact storage (MP4/WAV/JSON/PGN-equivalent for code runs).
- Clip generation tools (create evidence clips from timestamps).

Transcription, Diarization & Alignment

- Streaming ASR for live captions; batch ASR for polished transcripts.
- Word-level timestamps + confidence scores; silence and latency markers.
- Diarization and persistent speaker assignment when account/calendar data exists.
- Filler-word (ums/uhs), hedges, and disfluency tagging.

Hard-Skill Assessment (technical)

- Embedded code editor + sandboxed runner (Judge0 or containerized runner).
- Auto-grading harness (unit tests, output checks, performance/time limits).
- Multi-language support (Python, Java, JS, C++, SQL).
- Whiteboard/problem diagram capture and attachable artifacts.
- Plagiarism/code-similarity checks vs public repos + prior candidates.

Soft-Skill Assessment (multimodal)

- Linguistic analysis: coherence, structure (STAR detection), topical relevance.
- Vocal features: speaking rate, pitch variability, volume dynamics, monotony score.
- Response fluency: latency to respond, response completeness, tangential drift.
- Behavioral patterning: interruptions, turn-taking fairness, assertiveness markers.

Video & Non-verbal Analysis

- Face detection & ID (bounding boxes + tracklets).
- Head-pose (nods/shakes), gaze estimation (camera eye-contact), blink rates.
- Posture classification (upright vs slouched vs moving) and fidget detection.
- Facial expression / micro-emotion estimates (smile, confusion, frustration) with confidence.
- Environment checks: multiple faces, noisy background, prohibited items.

Anti-cheating / Proctoring

- Multi-face detection → flag extra people.
- Audio-video synchronization anomalies → secondary-device detection.
- Liveness checks: randomized challenge prompts (blink, head-turn).
- Deepfake and synthetic voice detection heuristics + human-review escalation.

- Environment & app checks (optional browser extension pre-check).
- IP/Geo anomalies and device fingerprinting; tamper-evident logs & hashes.

Scoring, Explainability & Review

- Per-question rubric scoring (auto + human override).
- Aggregated scores: hard-skill, soft-skill, proctor risk, hire recommendation.
- Evidence links: every score links to timestamped clips, code run logs, and transcript snippets.
- Reviewer workspace: annotate, correct, and queue for calibration.

Reports & Workflow Integrations

- Candidate report (strengths, areas to improve, clips).
- Recruiter dashboard with shortlist and clip-playlist.
- ATS Integrations: Greenhouse, Lever, Workable (webhooks/API).
- Notifications: Slack/Teams for results and follow-ups.
- Export: PDF, JSON; on-demand retention/erasures.

Platform & Governance

- Role-based access control, audit trails, retention policies, consent capture.
- Bias & fairness monitoring dashboards (statistical drift alerts).
- Human-in-loop review queues for flagged assessments.

5) Touchpoints and product surfaces

- **Candidate Portal (Web):** onboarding, consent, device test, interview play/record. FE features: simple, mobile-responsive interface, progress timer, inline help, lightweight camera/mic preview, transcript download.
- **Interviewer / Recruiter Dashboard (Web):** live view (optional) or post-interview reviewer UI; quick-scan summary cards, clip playlist, score breakdown, one-click ATS push. FE features: evidence scrubber (seek-to-clip), quick-rubric buttons, manual note fields.

- **Embedded Interview Widget:** embeddable iframe for partners/clients to drop into career sites.
 - **APIs & Webhooks:** create interviews, push results, fetch artifacts, ATS sync.
 - **Admin Console:** org settings, SSO, retention, seat management, audit logs.
 - **Mobile-lite Playback:** candidate and recruiter mobile apps or responsive web for playback & basic review.
- Design goal: FE is light and fast — the heavy lifting is server-side ML. Keep interactive FE features to a manageable set (recorder UI, transcript scrubber, clip creation, reviewer tools).

6) Technical Architecture Recommendations

Overarching approach

- Microservices-based backend; do heavy ML offline/batch where acceptable and streaming for live needs. Prioritize auditability: every automated decision stores evidence pointers.

Realtime / Media Ingest

- **WebRTC** for audio/video streaming; use an SFU like **mediasoup** or **Janus** for scaling multi-party media.
- Signaling via **FastAPI** + **Socket.IO** or WebSocket.

ASR & Speech

- Streaming baseline: **OpenAI Whisper (small/medium)** or **WhisperX** for alignment and timestamp quality. For lower-latency or on-premise, consider **Vosk** or custom small transformer.
- Post-process polishing with WhisperX and confidence-based re-transcription.

Diarization & Speaker Embeddings

- **pyannote.audio** for diarization and speaker-change detection. Use speaker embedding models from SpeechBrain or pyannote for persistent linking.

Video / Non-verbal Analysis

- **MediaPipe** (face mesh, pose, hands) as primary lightweight real-time library.
- **OpenCV** for frame ops and motion detection.
- Face embedding / identity/verification: **DeepFace** or **face-recognition** (dlib) for prototype; replace with hardened models as needed.
- Deepfake detection: fine-tune small classifiers on artifacts (research libs, FaceForensics++ style datasets) and keep thresholds conservative — always route to human review.

Prosody / Vocal Analysis

- **librosa + parselmouth (Praat bindings)** for pitch, jitter, shimmer, and prosody analysis.
- Extract short-term features and aggregate (wpm, pitch variance, energy).

NLP / Summarization / Scoring

- **Hugging Face Transformers** (instruction-tuned T5/BART/Llama-family) for summarization, STAR detection, and intent extraction.
- Use **sentence-transformers** for semantic search and candidate-to-job similarity.
- For explainability, use RAG pattern: retrieval of transcript snippets → LLM summarization constrained by evidence.

Code Execution & Hard-Skill Grading

- Sandboxed execution: **Judge0** for early builds; for scale, containerized sandbox execution with Kubernetes and strict resource limits (seccomp, cgroups).
- Unit-test-based grading harness with timeouts and memory caps. Store run artifacts and logs.

Anti-cheating & Security

- Liveness & challenge prompts implemented client-side (WebRTC capture & simple on-capture checks).
- Device fingerprinting + IP checks for anomaly detection.
- Cryptographic hashing (SHA-256) of recordings and logs for tamper-evidence.
- Optional browser extension for advanced environment checks (enterprise only).

Model infra & MLOps

- Training: **PyTorch** + **PyTorch Lightning**.
- Serving: **Triton Inference Server** for large models or **FastAPI** + **Uvicorn** for lighter endpoints.
- Feature store & experiment tracking: **MLflow** or **Weights & Biases**.
- Model versioning and rollout with canary + human-in-loop gating.

Data, Storage & Indexing

- Object storage: **S3** (or compatible).
- Metadata DB: **Postgres**.
- Queueing: **Redis** + **Celery** or **RabbitMQ** for background jobs.
- Vector DB: **FAISS** (self-host) or **Pinecone/Milvus** for semantic search and candidate embeddings.

Orchestration & Deployment

- Containerize (Docker); deploy on **Kubernetes** for scale (Helm charts). For early prototype, use managed k8s (GKE/EKS/AKS) or cloud run.
- CI/CD: GitHub Actions + ArgoCD for infra deployment.

UX / Frontend stack (minimal but professional)

- **React** + **TypeScript** (Next.js optional for SSR).
- UI library: Tailwind CSS + component kit (radix/shadcn if available).
- Media UI: react-webcam, custom WebRTC hooks; player using video.js or react-player.
- State management: React Query (TanStack) + Zustand for local UI state.

Edge capabilities & TTS

- For human-like assistant voice prompts, consider **EdgeTTS** or vendor TTS (Azure TTS / Amazon Polly Neural) with SSML. Keep TTS optional and configurable per org.

Orchestration of flows

- For complex decision flows (question branching, evidence linking), consider **LangGraph** or a light workflow engine (Temporal) to orchestrate LLM + ML services, ensuring traceability and replays.

Privacy, Compliance & Governance

- Implement SSO (SAML/OIDC), RBAC, data residency options. Use KMS for key management, and provide data retention & deletion APIs for GDPR/CCPA compliance. Keep an audit log store (immutable or append-only) for legal traceability.

Final implementation pragmatics

- AI students focus on ASR, video analysis, NLP scoring, code-grading, and anti-cheating models. Productionize the highest-impact models first (ASR + code sandbox + evidence linking).
- Prioritize explainability: every automated decision must point to a clip or test result — this reduces legal risk and increases trust.

SkillScreen is designed to be product-grade from day one: minimal but polished frontend, hardened ML back-end, explicit evidence trails, and conservative anti-cheating flows that defer to humans when needed.